



DPUSNDXEnvironment

Handles the playback of sound sources which have been placed on the track using Points of Interest (POI). Sound sources are placed at a fixed point along the track and audio output is adapted according to the relative position of the sound source to the simulation vehicle. Only one sound source is played at a time.

The sound samples which are included in the SILAB installation can easily be used by following the example at the end of this document.

This DPU is part of the default SILAB configuration. It can be found in the file DPUSNDX.inc.

1. Inputs

<i>Sources</i>	Set of names of the audio files which are loaded by the DPU.
<i>SoundPath</i>	Directory below „SILAB\lib\sounds\“ which is searched for the specified <i>Sources</i> . Default: “environment”
<i>Extension</i>	Filename extension of the audio files (<i>Sources</i>). Default: .wav
<i>POIName</i>	Family name of POIs which are considered by the DPU. Default: SNDX
<i>GlobalVolume</i>	Global volume for all sound outputs. Default: 1.0
<i>ManagedCursorIndex</i>	Index of the cursor which is used to search for relevant POIs. Default: 1
<i>HysteresisDistance</i>	Threshold to switch between two sound sources. Sound playback switches from source A to B if B is at least <i>HysteresisDistance</i> closer to the simulation vehicle than A [m]. Default: 1
<i>MaxDistance</i>	Maximum distance at which sound of a sound source is played [m]. Default: 250
<i>ReferenceDistance</i>	Reference distance for a sound source used for distance dependent volume adaptation [m]. Default: 5
<i>RollofFactor</i>	Additional factor for distance dependent volume adaptation. Default: 1

2. Functional principle

While the simulation is running, this DPU monitors all POIs with Family = *POIName* that are present in the current module. When *MaxDistance* to one of the POIs is undershot, its sound is played and continuously adapted depending on the relative position and distance to the simulation vehicle.

Each POI can play a different sound source and specify a volume for its audio.



3. Example

Using SILABAEEdit, an example configuration for a POI looks as shown in the image below. It specifies a POI which plays the sound sample "FewPeopleInTown" with a base volume of 0.5.

Name	Value
FewPeopleInTown	0.5

Figure 1 POI definition in SILABAEEdit

The DPU works well using the following example (default configuration in DPUSNDX.inc)

```
# Einbindung der DPUTRFDDataFull
DPUSNDXEnvironment SNDXEnv
{
    Computer = {SNDX};
    Index = 40;
    Sources = {"AmbientTown", "AmbientVillage", "Birds1", "Birds2",
    "Birds3", "Birds4", "BushCutting", "Construction", "FewPeopleInTown", "Few-
    PeopleOnField", "Fountain", "People1", "People2", "River"};
    ReferenceDistance = 7;
    RolloffFactor = 0.5;
};
```